

NextGen UCU Analytics – Business Model

1. Overview

NextGen UCU Analytics is a data warehouse and analytics platform tailored for Ugandan universities (starting with Uganda Christian University). It unifies academic, finance, and HR data into a single system that powers dashboards for Deans, HODs, Senate, HR, Finance, and Students, plus prediction models for performance and tuition risk.

This document summarises the business model in a way that can be shared as a PDF with stakeholders (VC, DVC, Deans, ICT, Finance, HR).

2. Customer Segments

- **Public & private universities in Uganda**
 - e.g. UCU, Makerere, Kyambogo, Mbarara, Ndejje, Uganda Martyrs, Kampala International.
 - Pain points: siloed systems (Academic Registrar, Finance, HR, Library), manual Excel reporting for NCHE/DIRECTORATE, late visibility into performance and fee collections.
 - **Faculties, Schools & Departments**
 - Deans and HODs who need faculty/department dashboards (students, lecturers, tuition, retention, FEX/MEX, high-school pipelines, HR analytics).
 - **Central university governance (Senate, Council, NCHE liaison)**
 - Senate members who want institution-wide dashboards for performance, completion, programme viability, and fee trends to support policy and accreditation.
 - **Operational directorates**
 - **Finance:** fee collection trends, ageing, programme profitability, high-risk students.
 - **HR:** staff workload, lecturer performance indicators, attendance, payroll analytics, part-time vs full-time ratios.
 - **ICT / MIS:** wants a standardised data platform rather than ad-hoc scripts and reports.
 - **Students (end-users)**
 - Self-service academic dashboards (grades, tuition, attendance, predictions), but they are **not payers**; they are part of the value proposition to the institution.
-

3. Value Proposition

For Universities (VC, DVC, Council)

- **Single source of truth** for academics, finance, and HR, built on a proper warehouse (star schema, ETL pipeline).

- **Regulatory reporting readiness** for NCHE and other bodies with consistent, auditable metrics.
- **Better resource allocation**: see which programmes, faculties, and campuses are under- or over-performing in real time.

For Faculties & Departments (Deans, HODs)

- **Role-based dashboards**:
 - Deans see only their faculty (students, lecturers, tuition, performance, retention, fee status).
 - HODs see only their department.
- **Granular student analytics**:
 - Performance by programme, course unit, gender, high-school, resident vs non-resident.
 - Early warning from FEX/MEX trends and predictive models.
- **Lecturer analytics**:
 - Performance indicators per lecturer / assistant lecturer, broken down by department, programme, gender, and now **full-time vs part-time**.

For HR

- **Complete staff directory** from HR + app users, with:
 - Senate members, Deans, HODs, Lecturers, Assistant Lecturers, Finance staff, HR staff, other employees.
 - Filters by faculty, department, and role group.
- **Attendance & payroll analytics**:
 - Attendance trends by rank (Senate, Deans, HODs, Lecturers, Assistant Lecturers, Finance, HR, Other).
 - Payroll totals and averages per rank, faculty, department.
- **Lecturer employment type**:
 - Breakdown of lecturers as **Full-time vs Part-time vs Other**, filtered by faculty and department.

For Finance

- **End-to-end tuition analytics**:
 - Payment distributions and trends by programme, faculty, gender, high-school, and year.
 - Identification of high-risk cohorts (e.g. certain programmes, high-schools, or years with chronic arrears).

For Students

- **Personalised academic cockpit**:
 - Only their own grades, trends over time, attendance, tuition status, and predictions.
 - Clear, friendly error messages and explanations (no raw backend errors).

4. Revenue Model

- **Institution-level annual subscription (SaaS licence)**

- Tiered by size (number of students / faculties):
 - Tier 1: small private universities & colleges.
 - Tier 2: mid-size universities.
 - Tier 3: large public universities.
 - Includes:
 - Access to the platform (dashboards + APIs).
 - Security, RBAC, and updates.
 - **Implementation & onboarding fees**
 - One-time fee per institution for:
 - Data mapping from existing SIS, finance, HR systems.
 - ETL setup and validation for UCU_SourceDB1/2 equivalents.
 - Dashboards customisation (branding, additional KPIs).
 - **Training & capacity building**
 - Paid training packages for:
 - Deans, HODs, Senate, Finance, HR.
 - ICT / MIS teams on ETL, data governance, and advanced analytics.
 - **Premium analytics & prediction add-ons**
 - Advanced machine learning models (drop-out prediction, fee default risk, scenario planning) sold as extra modules.
 - Custom reports for NCHE submissions, Council packs, or donor projects.
 - **Support & managed hosting**
 - Optional managed cloud hosting (e.g. on local data centre or cloud region compliant with Ugandan data rules).
 - SLA tiers (standard, premium 24/7) priced separately.
-

5. Cost Structure ("Constructor")

- **Core development & maintenance**
 - Software engineers (backend Flask, React frontend, ETL, data engineering).
 - Data scientists / ML engineers maintaining the prediction models.
 - **Infrastructure**
 - Databases (MySQL warehouse, RBAC DB).
 - Application servers and storage (cloud or on-prem).
 - Monitoring, backups, security tooling (firewalls, VPNs, SSL, IAM).
 - **Implementation & support**
 - Field engineers / implementation consultants for on-site visits at Ugandan campuses.
 - Support team handling tickets, upgrades, bug-fixes.
-

- **Sales, training, and partnerships**

- Account managers and pre-sales engineers.
 - Training materials, workshops, certification programmes.
 - Partnerships with:
 - Local ISPs / hosting providers.
 - Payment providers (MTN Mobile Money, Airtel Money, banks) where fee data integration is needed.
 - National IT bodies (e.g. NITA-U) for data governance alignment.
-

6. Channels & Customer Relationships

- **Direct sales to universities**

- ICT/MIS directorates, Academic Registrar, Finance, and VC's office.
- Pilots/POCs at one faculty (e.g. Business & Administration) before rollout.

- **Partner-led deployments**

- Collaboration with local software vendors and system integrators that already serve universities (SIS, ERP, LMS).

- **Ongoing relationships**

- Quarterly business reviews with data packs (key trends, risks, opportunities).
 - Joint innovation projects (new models, new KPIs) with early-adopter universities.
-

7. How to Use This Document as a PDF

- To produce a PDF for presentations or approvals:
 1. Open this file (**BUSINESS_MODEL.md**) in a markdown viewer (e.g. VS Code, a markdown preview plugin, or GitHub).
 2. Use "Print to PDF" or an export-to-PDF option from your editor or browser.
 3. Share the resulting PDF with university leadership, project supervisors, or potential partner institutions.